

Sample path dependent bounds for approximate information state

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Abstract—The approximate information state (AIS) framework of Subramanian et al (2022) provides sub-optimality bounds for structured policies in partially observable systems. These bounds are parameterized by worst-case approximation errors in per-step cost (ε_t) and AIS dynamics (δ_t) that hold uniformly over all histories. In many practical settings, the quality of approximation in cost and dynamics may vary significantly across histories. Consequently, the AIS suboptimality bounds that depend on worst-case approximations may turn out to be loose. In this paper, we derive *sample path dependent* sub-optimality bounds where the cost and dynamics approximation errors $\varepsilon_t(h_t, a_t)$ and $\delta_t(h_t, a_t)$ are allowed to depend on the realized history h_t and action a_t at time t . We illustrate through examples that these sample path dependent bounds provide tighter performance guarantees than the original AIS bounds.

I. INTRODUCTION

Partially observable Markov decision processes (POMDPs) provide a natural framework for sequential decision-making under incomplete state information. The standard approach for solving POMDPs converts them to fully observed MDPs over the space of posterior beliefs [1], [2]. However, this transformation is computationally challenging: finding optimal policies is PSPACE-hard [3] and the belief space is infinite-dimensional in general.

The *approximate information state* (AIS) framework [4] addresses this challenge by compressing the growing history into a fixed-dimensional representation that approximately satisfies two properties: (i) approximate sufficiency for predicting immediate rewards, and (ii) an approximate Markov property for predicting the next AIS value. Under these two conditions, the AIS framework provides explicit sub-optimality bounds for policies derived from an approximate dynamic program over the compressed representation. These bounds have been used to analyze a range of structured policies for POMDPs, including certainty equivalent policies [5], agent-state-based policies [6], self-predictive representation in Q-learning [7], new latent space representation learning for LQG control [8], worst case performance guarantees [9], video streaming with delayed feedback [10].

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The existing AIS bounds are parameterized by worst-case approximation errors: a reward prediction error $\{\varepsilon_t\}_{t \geq 1}$ and a transition prediction error $\{\delta_t\}_{t \geq 1}$, both of which must hold *uniformly over all histories and actions*. In practice, however, learned representations are often highly accurate for “typical” histories encountered during normal operation but may perform poorly on rare or adversarial ones. The worst-case formulation cannot distinguish between these cases, leading to bounds that can be significantly loose.

Contributions: In this paper, we address this limitation by developing *sample path dependent* AIS bounds. Our specific contributions are:

- We derive sub-optimality bounds for AIS-based policies in the single-agent (centralized) setting where the approximation errors $\varepsilon_t(h_t, a_t)$ and $\delta_t(h_t, a_t)$ are allowed to depend on both the realized history and the action. The error is characterized through an α - β recursion: $\beta_t(h_t, a_t)$ bounds the Q-function approximation error for a specific history-action pair, and $\alpha_t(h_t)$ bounds the value function approximation error. The final sub-optimality bound replaces the worst-case α of the original AIS theory with $2\alpha_t(h_t)$ pointwise, or $2\mathbb{E}^\pi[\alpha_t(H_t)]$ in expectation.
- We provide a characterization of $\alpha_t(h_t)$ that is tighter than the naive supremum over actions: $\alpha_t(h_t) = \max\{\beta_t(h_t, \pi_t^*(h_t)), \beta_t(h_t, \hat{\pi}_t(\sigma_t(h_t)))\}$, and discuss how the action set can potentially be pruned to obtain even tighter bounds.

The rest of the paper is organized as follows. We present the system model and review the AIS framework in Sec. II. We present the sample path dependent bounds for the centralized setting in Sec. III. We conclude in Sec. V.

Notation: We use uppercase letters to denote random variables (e.g., S , A , etc.), the corresponding lowercase letters to denote their realizations (e.g., s , a , etc.), and the corresponding calligraphic letters to denote their space of realizations (e.g., $\mathcal{S}$, $\mathcal{A}$, etc.). Subscripts denote time, so S_t denotes a variable at time t . The notation $S_{1:t}$ is a shorthand for the sequence $(S_1, \dots, S_t)$. We use $\mathbb{R}$ to denote the set of real numbers. For a topological space $\mathcal{X}$, $\Delta(\mathcal{X})$ denotes the set of all probability measures on $\mathcal{X}$ and $\mathcal{B}(\mathcal{X})$ denotes the set of all bounded and measurable real-valued functions on $\mathcal{X}$. We use $\mathbb{P}(\cdot)$ to denote the probability of an event and $\mathbb{E}[\cdot]$ to denote the expectation of a random variable.

II. SYSTEM MODEL AND BACKGROUND

A. System model

Consider a partially observable Markov decision process (POMDP), denoted by $\mathcal{P}$, that runs for a finite horizon T .

At each time $t \in \{1, \dots, T\}$, the system has a state $S_t \in \mathcal{S}$, the controller receives an observation $Y_t \in \mathcal{Y}$, and selects an action $A_t \in \mathcal{A}$. The initial state and observation (S_1, Y_1) are distributed according to $\xi \in \Delta(\mathcal{S} \times \mathcal{Y})$. The system dynamics are Markovian: there exist stochastic kernels $P_t: \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S} \times \mathcal{Y})$, $t \in \{1, \dots, T-1\}$, such that for any measurable subsets M_S and M_Y of $\Delta(\mathcal{S})$ and $\Delta(\mathcal{Y})$, we have

$$\begin{aligned} \mathbb{P}(S_{t+1} \in M_S, Y_{t+1} \in M_Y | s_{1:t}, y_{1:t}, a_{1:t}) \\ = P_t(M_S, M_Y | s_t, a_t). \end{aligned}$$

The controller has access to the history $h_t = (y_{1:t}, a_{1:t-1})$ at time t . Let $\mathcal{H}_t$ denote the space of realizations of h_t . At each time t , the system incurs a per-step cost $c_t(h_t, a_t)$ where $c_t: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ is a measurable function.

A policy $\pi = (\pi_1, \dots, \pi_T)$, $\pi_t: \mathcal{H}_t \rightarrow \mathcal{A}$, is a collection of measurable functions that map histories to actions. The value function of policy π is

$$V_t^\pi(h_t) = \mathbb{E}^\pi \left[\sum_{\tau=t}^T c_\tau(H_\tau, A_\tau) \mid h_t \right].$$

The *optimal* value function is defined as

$$V_t^*(h_t) = \inf_{\pi} V_t^\pi(h_t), \quad t \in \{1, \dots, T\} \quad (1)$$

where the infimum is over all history dependent policies.

The value function of a policy can be recursively computed as follows. Define $V_{T+1}^\pi(h_{T+1}) = 0$ for all $h_{T+1} \in \mathcal{H}_{T+1}$ and then for $t \in \{T, \dots, 1\}$ recursively define

$$\begin{aligned} Q_t^\pi(h_t, a_t) &= c_t(h_t, a_t) + \int P_t(dy_{t+1} | h_t, a_t) V_{t+1}^\pi(h_t, a_t, y_{t+1}), \\ V_t^\pi(h_t) &= Q_t^\pi(h_t, \pi_t(h_t)). \end{aligned}$$

Remark 1 Typically, POMDPs models assume that the per-step cost depends on the current state instead of the history of observations. Such models with a per-step cost $k_t(S_t, A_t)$ may be viewed as a special case of our model by taking

$$c_t(h_t, a_t) = \mathbb{E}[k_t(S_t, A_t) | H_t = h_t, A_t = a_t].$$

Assumption 1 (DP solvability) The POMDP $\mathcal{P}$ satisfies the following conditions all $t \in \{1, \dots, T\}$,

- 1) $V_t^*: \mathcal{H}_t \rightarrow \mathbb{R}$ defined in (1) is a measurable function.
- 2) $Q_t^*: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ defined as

$$\begin{aligned} Q_t^*(h_t, a_t) &= c_t(h_t, a_t) \\ &+ \int P_t(dy_{t+1} | h_t, a_t) V_{t+1}^*(h_t, a_t, y_{t+1}) \end{aligned}$$

is a measurable function where $V_{T+1}^*(h_{T+1}) \equiv 0$.

- 3) V_t^* and Q_t^* are related by

$$V_t^*(h_t) = \inf_{a_t \in \mathcal{A}} Q_t^*(h_t, a_t).$$

- 4) There exists a policy π^* such that for all t and all h_t , $V_t^{\pi^*}(h_t) = V_t^*(h_t)$.

We refer to these conditions as DP solvability (short for dynamic programming solvability).

B. Background on integral probability metrics (IPMs)

Definition 1 ([11]) Let $\mathfrak{F}$ be a convex and balanced subset of bounded measurable functions from a metric space $(\mathcal{Z}, d_{\mathcal{Z}})$ to $\mathbb{R}$. Then, the IPM distance (w.r.t. $\mathfrak{F}$) between two probability laws ν_1, ν_2 on $\mathcal{Z}$ is given by

$$d_{\mathfrak{F}}(\nu_1, \nu_2) = \sup_{f \in \mathfrak{F}} \left| \int f d\nu_1 - \int f d\nu_2 \right|.$$

Definition 2 In the setting of Definition 1, the Minkowski functional of any bounded measurable function $f: \mathcal{Z} \rightarrow \mathbb{R}$ is defined as

$$\rho_{\mathfrak{F}}(f) = \inf \left\{ \rho \in \mathbb{R}_{>0} : \frac{f}{\rho} \in \mathfrak{F} \right\}.$$

Note that if for every positive ρ , $f/\rho \notin \mathfrak{F}$, then $\rho_{\mathfrak{F}}(f) = \infty$.

An immediate consequence of the above two definitions is that for any bounded measurable function $f: \mathcal{Z} \rightarrow \mathbb{R}$,

$$\left| \int f d\nu_1 - \int f d\nu_2 \right| \leq \rho_{\mathfrak{F}}(f) d_{\mathfrak{F}}(\nu_1, \nu_2). \quad (2)$$

Examples of IPMs include (i) the total variation distance with the corresponding Minkowski functional being the half of the span of the function, and (ii) the Wasserstein-1 distance with the corresponding Minkowski functional being the Lipschitz constant of the function.

C. Approximate information state: review

Definition 3 Given a metric space $(\mathcal{Z}, d_{\mathcal{Z}})$, an approximate information state (AIS) generator for a POMDP $\mathcal{P}$ consists of:

- a sequence of measurable history compression functions $\sigma_t: \mathcal{H}_t \rightarrow \mathcal{Z}$, $t \in \{1, \dots, T\}$
- a sequence of bounded measurable cost approximation functions $\hat{c}_t: \mathcal{Z} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$, $t \in \{1, \dots, T\}$
- a sequence of stochastic kernels $\hat{P}_t: \mathcal{Z} \times \mathcal{A} \rightarrow \Delta(\mathcal{Z})$, $t \in \{1, \dots, T\}$

that satisfy the following property (referred to as measurable selection): for every measurable function $\hat{V}: \mathcal{Z} \rightarrow \mathbb{R}$ and each time $t \in \{1, \dots, T\}$, there exists a measurable selector $\hat{\pi}: \mathcal{Z} \rightarrow \mathcal{A}$ such that

$$\begin{aligned} \inf_{a \in \mathcal{A}} \left\{ \hat{c}_t(z, a) + \int_{\mathcal{Z}} \hat{P}_t(dz' | z, a) \hat{V}(z') \right\} \\ = \hat{c}_t(z, \hat{\pi}(z)) + \int_{\mathcal{Z}} \hat{P}_t(dz' | z, \hat{\pi}(z)) \hat{V}(z') =: \hat{V}_+(z), \end{aligned}$$

and $\hat{V}_+: \mathcal{Z} \rightarrow \mathbb{R}$ defined above is a measurable function.

Essentially, an AIS defines a Markov decision process (MDP) with state space $\mathcal{Z}$, action space $\mathcal{A}$, per-step cost $\{\hat{c}_t\}_{t=1}^T$, and dynamics $\{\hat{P}_t\}_{t=1}^T$. An implication of the measurable selection assumption is that the optimal policy of the MDP is given by the following dynamic program. Define $\hat{V}_{T+1}^*(z) \equiv 0$ and for $t \in \{T, \dots, 1\}$, recursively define

$$\hat{Q}_t^*(z_t, a_t) = \hat{c}_t(z_t, a_t) + \int \hat{P}_t(dz_{t+1} | z_t, a_t) \hat{V}_{t+1}^*(z_{t+1}),$$

$$\hat{V}_t^*(z_t) = \inf_{a_t \in \mathcal{A}} \hat{Q}_t^*(z_t, a_t).$$

Let $\hat{\pi}^* = (\hat{\pi}_1^*, \dots, \hat{\pi}_T^*)$, $\hat{\pi}_t^*: \mathcal{Z} \rightarrow \mathcal{A}$, be any policy such that $\hat{\pi}_t^*(z_t)$ is a minimizer of $\hat{Q}_t(z_t, a_t)$.

The policy $\hat{\pi}^*$ yields a feasible policy $\pi = (\pi_1, \dots, \pi_T)$ for the original POMDP, where

$$\pi_t = \hat{\pi}_t^* \circ \sigma_t, \quad \text{i.e., } \pi_t(h_t) = \hat{\pi}_t^*(\sigma_t(h_t)). \quad (3)$$

The AIS bound of [4] provides an upper bound on the sub-optimality gap of π .

Theorem 1 ([4]) Suppose for all $h_t \in \mathcal{H}_t$ and $a \in \mathcal{A}$, the AIS generator satisfies

$$|c_t(h_t, a) - \hat{c}_t(\sigma_t(h_t), a)| \leq \bar{\epsilon}_t \quad (\text{AP1})$$

and

$$d_{\mathcal{F}}(\nu_t(\cdot|h_t, a), \hat{P}_t(\cdot|\sigma_t(h_t), a)) \leq \bar{\delta}_t \quad (\text{AP2})$$

where $\nu_t(\cdot|h_t, a) = \mathbb{P}(\sigma_{t+1}(H_{t+1}) \in \cdot | h_t, a)$ is the true conditional distribution of $\sigma_{t+1}(H_{t+1})$ given (h_t, a) in model $\mathcal{P}$. Then

$$|V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t))| \leq \bar{\alpha}_t \quad (4)$$

and

$$V_t^\pi(h_t) - V_t^*(h_t) \leq 2\bar{\alpha}_t, \quad (5)$$

where $\bar{\alpha}_T = \bar{\epsilon}_T$ and for $t < T$,

$$\bar{\alpha}_t = \bar{\epsilon}_t + \sum_{\tau=t}^{T-1} [\bar{\delta}_\tau \rho_{\mathcal{F}}(\hat{V}_{\tau+1}) + \bar{\epsilon}_{\tau+1}] = \bar{\epsilon}_t + \bar{\delta}_t \rho_{\mathcal{F}}(\hat{V}_{t+1}) + \bar{\alpha}_{t+1}.$$

Note that the bound (5) holds pointwise for every history h_t and involves worst-case errors $\bar{\epsilon}_t, \bar{\delta}_t$ that are independent of the history. The goal of this paper is to tighten this bound by allowing the errors to depend on the specific history and action encountered.

III. SAMPLE PATH DEPENDENT BOUNDS

We replace the uniform AIS conditions with history-and-action-dependent versions.

Assumption 2 For each $t \in \{1, \dots, T\}$, there exists a measurable function $\epsilon_t: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $h_t \in \mathcal{H}_t$ and $a \in \mathcal{A}$,

$$|c_t(h_t, a) - \hat{c}_t(\sigma_t(h_t), a)| \leq \epsilon_t(h_t, a). \quad (6)$$

Assumption 3 For each $t \in \{1, \dots, T-1\}$, there exists a measurable function $\Delta_t: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $h_t \in \mathcal{H}_t$ and $a \in \mathcal{A}$,

$$\left| \int \hat{V}_{t+1}^*(z_{t+1}) \nu_t(dz_{t+1} | h_t, a) - \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}_t(dz_{t+1} | \sigma(h_t), a) \right| \leq \Delta_t(h_t, a).$$

A sufficient condition for Assumption 3 is as follows.

Assumption 4 For each $t \in \{1, \dots, T-1\}$, there exists a measurable function $\delta_t: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $h_t \in \mathcal{H}_t$ and $a \in \mathcal{A}$,

$$d_{\mathcal{F}}(\nu_t(\cdot|h_t, a), \hat{P}_t(\cdot|\sigma_t(h_t), a)) \leq \delta_t(h_t, a) \quad (7)$$

Under Assumption 4, it is easy to check that Assumption 3 holds with $\Delta_t(h_t, a) = \rho_{\mathcal{F}}(\hat{V}_{t+1}^*) \delta_t(h_t, a)$ by IPM property (2).

Theorem 2 Suppose that for each $t \in \{1, \dots, T\}$, there exist measurable functions $\alpha_t: \mathcal{H}_t \rightarrow \mathbb{R}$ and $\beta_t: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ such that $\beta_T(h_T, a_T) \geq \epsilon_T(h_T, a_T)$ and for $t < T$:

$$\beta_t(h_t, a_t) \geq \epsilon_t(h_t, a_t) + \int \alpha_{t+1}(h_t, a_t, y_{t+1}) P(dy_{t+1} | h_t, a_t) + \Delta_t(h_t, a_t), \quad (8)$$

and for all $t \in \{1, \dots, T\}$

$$\alpha_t(h_t) \geq \max\{\beta_t(h_t, \pi_t^*(h_t)), \beta_t(h_t, \hat{\pi}_t^*(\sigma_t(h_t)))\}. \quad (9)$$

Then, under Assumptions 2 and 3, for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, the AIS-based policy $\pi_t(h_t) = \hat{\pi}_t^*(\sigma_t(h_t))$ satisfies

$$V_t^\pi(h_t) - V_t^*(h_t) \leq 2\alpha_t(h_t). \quad (10)$$

See the proof of the theorem in the Appendix

Corollary 1 Under Assumptions 2 and 4, the result of Theorem 2 remains true with $\Delta_t(h_t, a_t)$ in (8) replaced by $\rho_{\mathcal{F}}(\hat{V}_{t+1}) \delta_t(h_t, a_t)$.

Remark 2 If (AP1) and (AP2) of Theorem 1 hold, then Assumptions 2 and 4 hold with $\epsilon_t(h_t, a_t) = \bar{\epsilon}_t$ and $\delta_t(h_t, a_t) = \bar{\delta}_t$. In this case, the functions α_t and β_t of Theorem 2 can be defined as constants: $\beta_t(h_t, a_t) = \alpha_t(h_t) = \bar{\alpha}_t$ where $\bar{\alpha}_t$ are defined as in Theorem 1. With these definitions, the sub-optimality bound in (10) of Theorem 2 becomes identical to the bound in (5) of Theorem 1.

A. Simplification of the upper bound

The sub-optimality bounds of Theorem 2 are in terms of the functions $\alpha_t(h_t)$ and $\beta_t(h_t, a_t)$ which are defined in a backward manner. As we will show in Section IV, it may be possible in some cases to construct functions $\alpha_t(h_t)$ and $\beta_t(h_t, a_t)$ using properties of the model and the AIS generator. However, in general, finding α_t, β_t functions that satisfy the conditions of Theorem 2 can be computationally challenging and/or impractical. Two key sources of this difficulty are: (i) the integral in (8) of $\alpha_{t+1}(\cdot)$ with respect to the probability distribution of the next observation given the current history and action may be hard to evaluate, especially if the model $\mathcal{P}$ is not exactly known or is too complex; (ii) the maximization in (9) involves the optimal action $\pi_t^*(h_t)$ in $\mathcal{P}$ which is not known. In this subsection, we outline a few approaches to simplify the computation of $\alpha_t(h_t)$ and $\beta_t(h_t, a_t)$ which in turn lead to computable sub-optimality bounds.

1) *Suprema based α, β* : We can bypass the integral in (8) and the maximization in (9) by replacing them with suprema over appropriate spaces as described in the following corollary.

Corollary 2 For all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, define $\alpha_t(h_t)$ and $\beta_t(h_t, a_t)$ as follows: $\beta_T(h_T, a_T) = \epsilon_T(h_T, a_T)$ and for $t < T$:

$$\beta_t(h_t, a_t) = \epsilon_t(h_t, a_t) + \sup_{y \in \mathcal{Y}} \alpha_{t+1}(h_t, a_t, y) + \Delta_t(h_t, a_t), \quad (11)$$

and for all $t \in \{1, \dots, T\}$

$$\alpha_t(h_t) = \sup_{a \in \mathcal{A}(h_t)} \beta_t(h_t, a), \quad (12)$$

where $\mathcal{A}(h_t)$ is any subset of $\mathcal{A}$ that is guaranteed to contain $\pi_t^*(h_t)$ and $\hat{\pi}_t^*(\sigma_t(h_t))$. If the functions $\alpha_t(\cdot)$ and $\beta_t(\cdot)$ are measurable, then

$$V_t^\pi(h_t) - V_t^*(h_t) \leq 2\alpha_t(h_t).$$

PROOF The proof follows from the fact that the functions $\alpha_t(\cdot)$ and $\beta_t(\cdot)$ defined in the Corollary satisfy the conditions of Theorem 2. ■

The subset $\mathcal{A}(h_t)$ in Corollary 2 can be obtained from properties of the optimal policy. For example, if it is known that π^* and $\hat{\pi}^*$ are “bang-bang” policies [12], then $\mathcal{A}(h_t)$ can be the set of extreme points of $\mathcal{A}$. Another choice of $\mathcal{A}(h_t)$ is given by

$$\mathcal{A}(h_t) := \left\{ a \in \mathcal{A} : \hat{Q}_t^*(\sigma_t(h_t), a) - \beta_t(h_t, a) \leq \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) + \beta_t(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \right\}. \quad (13)$$

Using Lemma 2 in the appendix, it can be verified that $\pi_t^*(h_t)$ and $\hat{\pi}_t^*(\sigma_t(h_t))$ belong to $\mathcal{A}(h_t)$ defined above.

2) α, β as functions of $\sigma_t(h_t)$: The result of Theorem 2 can be simplified if in Assumptions 2-4, the $\epsilon_t, \Delta_t, \delta_t$ depend on $\sigma_t(h_t)$ rather than directly on h_t (e.g. the RHS of (6) is $\epsilon_t(\sigma_t(h_t), a)$ instead of $\epsilon_t(h_t, a)$). In this case, the following corollary can be obtained using arguments similar to those used in Theorem 2 and Corollary 2.

Corollary 3 For all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, define $\alpha_t(\sigma_t(h_t))$ and $\beta_t(\sigma_t(h_t), a_t)$ as follows: $\beta_T(\sigma_T(h_T), a_T) = \varepsilon_T(\sigma_T(h_T), a_T)$ and for $t < T$:

$$\begin{aligned} \beta_t(\sigma_t(h_t), a_t) &= \varepsilon_t(\sigma_t(h_t), a_t) + \sup_{y \in \mathcal{Y}} \alpha_{t+1}(\sigma_t(h_t), a_t, y) \\ &\quad + \Delta_t(\sigma_t(h_t), a_t), \end{aligned} \quad (14)$$

and for all $t \in \{1, \dots, T\}$

$$\alpha_t(\sigma_t(h_t)) = \sup_{a \in \mathcal{A}} \beta_t(\sigma_t(h_t), a). \quad (15)$$

If the functions $\alpha_t(\cdot)$ and $\beta_t(\cdot)$ are measurable, then

$$V_t^\pi(h_t) - V_t^*(h_t) \leq 2\alpha_t(\sigma_t(h_t)).$$

3) Computing α, β using a fictitious control problem: Consider a fictitious control problem with the same POMDP model $\mathcal{P}$ as in Section II but with a per-step reward $r_t(h_t, a_t)$ (instead of a per-step cost) defined as follows:

$$r_t(h_t, a_t) = \varepsilon_t(h_t, a_t) + \Delta_t(h_t, a_t). \quad (16)$$

We will refer to the fictitious control problem described above as **Problem FP**.

As in Section II, a policy $\pi = (\pi_1, \dots, \pi_T)$, $\pi_t: \mathcal{H}_t \rightarrow \mathcal{A}$, is a collection of measurable functions that map histories to actions. The value function of policy π in the fictitious control problem is

$$V_t^{\pi, \text{FP}}(h_t) = \mathbb{E}^\pi \left[\sum_{\tau=t}^T r_\tau(H_\tau, A_\tau) \mid h_t \right].$$

The *optimal* value function for the fictitious control problem is defined as

$$V_t^{*, \text{FP}}(h_t) = \sup_{\pi} V_t^{\pi, \text{FP}}(h_t), \quad t \in \{1, \dots, T\} \quad (17)$$

where the supremum is over all history dependent policies. We assume that Problem FP satisfies a DP solvability condition analogous to Assumption 1.

Assumption 5 (DP solvability for Problem FP) Problem FP satisfies the following conditions all $t \in \{1, \dots, T\}$,

- 1) $V_t^{*, \text{FP}}: \mathcal{H}_t \rightarrow \mathbb{R}$ defined in (17) is a measurable function.
- 2) $Q_t^{*, \text{FP}}: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ defined as

$$\begin{aligned} Q_t^{*, \text{FP}}(h_t, a_t) &= r_t(h_t, a_t) \\ &\quad + \int P_t(dy_{t+1} | s_t, a_t) V_{t+1}^{*, \text{FP}}(h_t, a_t, y_{t+1}) \end{aligned}$$

is a measurable function where $V_{T+1}^{*, \text{FP}}(h_{T+1}) \equiv 0$.

- 3) $V_t^{*, \text{FP}}$ and $Q_t^{*, \text{FP}}$ are related by

$$V_t^{*, \text{FP}}(h_t) = \sup_{a_t \in \mathcal{A}} Q_t^{*, \text{FP}}(h_t, a_t).$$

We have the following lemma.

Lemma 1 For all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, define $\alpha_t(h_t)$ and $\beta_t(h_t, a_t)$ as follows:

$$\beta_t(h_t, a_t) = Q_t^{*, \text{FP}}(h_t, a_t), \quad \alpha_t(h_t) = V_t^{*, \text{FP}}(h_t). \quad (18)$$

Then, these $\alpha_t, \beta_t, t = 1, \dots, T$ satisfy the conditions of Theorem 2. Consequently,

$$V_t^\pi(h_t) - V_t^*(h_t) \leq 2\alpha_t(h_t) = 2V_t^{*, \text{FP}}(h_t). \quad (19)$$

Lemma 1 suggests that if we can find or over-approximate the optimal value function of Problem FP, we have a sub-optimality bound for the AIS-based policy π . The derivation has three explicit steps: obtain the FP reduction (19), upper bound the FP value using an AIS model (20), and combine them into a policy sub-optimality bound (21). One way to find an over-approximation of $V_t^{*, \text{FP}}(h_t)$ is to use an AIS-generator for Problem FP, as described in Section II-C.

Let $\tilde{\sigma}_{1:T}, \tilde{P}_{1:T}, \tilde{r}_{1:T}$ be an AIS generator for Problem FP (see Definition 3 with inf replaced by sup). Assume that this AIS generator satisfies (AP1) and (AP2) of Theorem 1 (with c_t replaced by r_t and $\hat{c}_t$ by $\tilde{r}_t$). As discussed in Section II-C, an AIS generator defines an MDP. Let $\hat{V}^{*, \text{FP}}$ be the optimal value functions of this AIS MDP. Then, from (4) in Theorem 1, we have that

$$V_t^{*, \text{FP}}(h_t) \leq \hat{V}_t^{*, \text{FP}}(\tilde{\sigma}(h_t)) + \bar{\alpha}_t. \quad (20)$$

Combining (19) and (20), we get the following sub-optimality bound

$$V_t^\pi(h_t) - V_t^*(h_t) \leq 2V_t^{*, \text{FP}}(h_t) \leq 2\hat{V}_t^{*, \text{FP}}(\tilde{\sigma}(h_t)) + 2\bar{\alpha}_t. \quad (21)$$

In other words, the sub-optimality of the policy generated using AIS generator $\sigma, \hat{P}, \hat{c}$ can be upper bounded using the optimal value function of the MDP corresponding to the AIS generator $\tilde{\sigma}, \tilde{P}, \tilde{r}$ for Problem FP.

IV. ILLUSTRATIVE EXAMPLES

A. MDP Model Approximation

In this subsection, we specialize our main result to the fully observed setting, where the POMDP reduces to a Markov decision process (MDP) and the state s_t serves as a sufficient statistic for the history h_t .

Consider a finite-horizon MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \{P_t\}_{t=1}^{T-1}, \{c_t\}_{t=1}^T, T \rangle$ and an approximate MDP $\hat{\mathcal{M}} = \langle \mathcal{S}, \mathcal{A}, \{\hat{P}_t\}_{t=1}^{T-1}, \{\hat{c}_t\}_{t=1}^T, T \rangle$ on the same state and action spaces. In the MDP setting, the observation equals the state ($Y_t = S_t$) and the AIS history compression is the identity on the current state: $\sigma_t(h_t) = s_t$ and $\mathcal{Z} = \mathcal{S}$.

We make the following assumptions:

Assumption 6 For each $t \in \{1, \dots, T\}$, there exists a measurable function $\varepsilon_t: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,

$$|c_t(s, a) - \hat{c}_t(s, a)| \leq \varepsilon_t(s, a). \quad (22)$$

Assumption 7 For each $t \in \{1, \dots, T-1\}$, there exists a measurable function $\Delta_t: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,

$$\left| \int \hat{V}_{t+1}^*(s') P_t(ds'|s, a) - \int \hat{V}_{t+1}^*(s') \hat{P}_t(ds'|s, a) \right| \leq \Delta_t(s, a). \quad (23)$$

A sufficient condition for Assumption 7 is the following.

Assumption 8 For each $t \in \{1, \dots, T-1\}$, there exists a measurable function $\delta_t: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,

$$d_{\mathcal{F}}(P_t(\cdot|s, a), \hat{P}_t(\cdot|s, a)) \leq \delta_t(s, a). \quad (24)$$

Under Assumption 8, Assumption 7 holds with $\Delta_t(s, a) = \rho_{\mathcal{F}}(\hat{V}_{t+1}^*) \cdot \delta_t(s, a)$.

Remark 3 In the MDP setting with $\sigma_t(h_t) = s_t$, Assumption 6 implies Assumption 2 as discussed in Remark 1. Similarly, Assumption 7 implies Assumption 3 and Assumption 8 implies Assumption 4, with the distribution $\nu_t(\cdot|h_t, a) = P_t(\cdot|s_t, a)$.

We can now state the MDP specialization of the result in Theorem 2, which may be viewed as a generalization of the approximation bounds of [13].

Corollary 4 Consider two finite-horizon MDPs $\mathcal{M}$ and $\hat{\mathcal{M}}$ on the same state and action spaces, both satisfying Assumption 1.

Under Assumptions 6 and 7, suppose that for each $t \in \{1, \dots, T\}$, there exist measurable functions $\alpha_t: \mathcal{S} \rightarrow \mathbb{R}$ and $\beta_t: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ such that $\beta_T(s, a) \geq \varepsilon_T(s, a)$ and for $t < T$:

$$\beta_t(s, a) \geq \varepsilon_t(s, a) + \int \alpha_{t+1}(s') P_t(ds'|s, a) + \Delta_t(s, a), \quad (25)$$

and for all $t \in \{1, \dots, T\}$:

$$\alpha_t(s) \geq \max\{\beta_t(s, \pi_t^*(s)), \beta_t(s, \hat{\pi}_t^*(s))\}. \quad (26)$$

Then, the policy $\pi_t(s) = \hat{\pi}_t^*(s)$ satisfies for all t and s :

$$|V_t^\pi(s) - V_t^*(s)| \leq 2\alpha_t(s). \quad (27)$$

PROOF Take $Y_t = S_t$, $\sigma_t(h_t) = s_t$, and $\mathcal{Z} = \mathcal{S}$ in Theorem 2. Under full state observation, the expected cost $\int c_t(s_t, a) P(ds_t|h_t) = c_t(s_t, a)$, so Assumption 2 is satisfied with $\varepsilon_t(h_t, a) = \varepsilon_t(s_t, a)$ under Assumption 6. Similarly, Assumption 3 is satisfied with $\Delta_t(h_t, a) = \Delta_t(s_t, a)$ under Assumption 7. The integral $\int \alpha_{t+1}(h_t, a_t, y_{t+1}) P(dy_{t+1}|h_t, a_t)$ becomes $\int \alpha_{t+1}(s') P_t(ds'|s, a)$. The conditions of Theorem 2 are then satisfied with the state-dependent functions α_t and β_t specified above. ■

B. Approximate Inventory Management

We illustrate the results of Corollary 4 on an inventory management problem. Consider an inventory control MDP with state space $\mathcal{S} = \{-S_{\max}, \dots, S_{\max}\}$, action space $\mathcal{A} = \{0, 1, \dots, S_{\max}\}$, and dynamics

$$S_{t+1} = [S_t + A_t - N_t]_{-S_{\max}}^{S_{\max}}$$

where $[\cdot]_{-S_{\max}}^{S_{\max}}$ clips its argument to $[-S_{\max}, S_{\max}]$ and the demand N_t is an i.i.d. Binomial(n, q) process. Let the per-step cost be $c(s, a) = pa + h(s)$, where

$$h(s) = \begin{cases} c_h s & s \geq 0, \\ -c_s s & s < 0, \end{cases}$$

p is the per-unit ordering cost, c_h is the holding cost, and c_s is the shortage cost. The system runs for an infinite horizon and future is discounted by a factor $\gamma \in (0, 1)$. We parameterize such a model as $\mathcal{M} = (S_{\max}, \gamma, n, q, c_h, c_s, p)$.

To apply Corollary 4 directly, we reformulate the problem as a finite-horizon MDP with horizon T (for some large T) by absorbing the discount factor into a time-varying cost $c_t(s, a) = \gamma^{t-1}[pa + h(s)]$ for $t = 1, \dots, T$, and similarly for the approximate model. The transitions remain stationary: $P_t = P$ and $\hat{P}_t = \hat{P}$ for all t .

We consider the following two models:

- True model $\mathcal{M} = (500, 0.75, 10, 0.4, 4.0, 2, 5)$.
- Approx. model $\hat{\mathcal{M}} = (500, 0.75, 10, 0.5, 3.8, 2, 5)$.

Furthermore, Assumption 6 is satisfied with $\varepsilon_t(s, a) = \gamma^{t-1}\varepsilon(s)$, where $\varepsilon(s) = |h_{\mathcal{M}}(s) - h_{\hat{\mathcal{M}}}(s)|$ depends only on the state since the ordering cost p is identical in both models. Explicitly,

$$\varepsilon(s) = \begin{cases} |c_h - \hat{c}_h| \cdot s = 0.2s & s \geq 0, \\ |c_s - \hat{c}_s| \cdot |s| = 0 & s < 0. \end{cases}$$

Also, Assumption 7 is satisfied with

$$\Delta_t(s, a) = \left| \sum_{s'} \hat{V}_{t+1}^*(s') [P(s'|s, a) - \hat{P}(s'|s, a)] \right|,$$

where $\hat{V}_{t+1}^*$ is the time- $(t+1)$ optimal value function of $\hat{\mathcal{M}}$, computed by backward induction. Both ε_t and Δ_t can be computed exactly since the state and action spaces are finite.

We define α, β functions satisfying Corollary 4 as follows. At the terminal time, $\beta_T(s, a) = \gamma^{T-1}\varepsilon(s)$ and $\alpha_T(s) = \gamma^{T-1}\varepsilon(s)$. For $t < T$:

$$\begin{aligned} \beta_t(s, a) &= \gamma^{t-1}\varepsilon(s) + \sum_{s'} \alpha_{t+1}(s') P(s'|s, a) + \Delta_t(s, a) \\ \alpha_t(s) &= \max_{a \in \mathcal{A}(s)} \beta_t(s, a), \end{aligned} \quad (28)$$

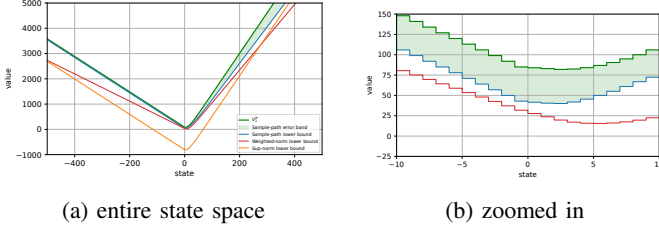


Fig. 1: Comparison of lower bounds on $V_1^*(s)$ for the finite-horizon inventory management model.

where $A(s)$ is the action set we will determine below.

Corollary 4 then implies

$$V_1^{\hat{\pi}^*}(s) - 2\alpha_1(s) \leq V_1^*(s) \leq V_1^{\hat{\pi}^*}(s). \quad (29)$$

As given in (26), the tightest bound uses

$$A(s) = \{\pi_t^*(s), \hat{\pi}_t^*(s)\}, \quad (30)$$

which requires the true model's optimal policy. Since the optimal policy is base-stock with level¹ $s_t^* \leq n = 10$, the restricted action set

$$A_{\text{res}}(s) = \{\max(0, s^* - s) : s^* \in \{0, \dots, n\}\} \cup \{\hat{\pi}_t^*(s)\} \quad (31)$$

contains at most $n + 2 = 12$ actions and includes $\pi_t^*(s)$ without requiring its explicit computation. This yields a bound nearly as tight as the two-action variant while requiring only structural knowledge of the policy class. We use $2\alpha_t^{\max}(s)$ and $2\alpha_1^{\text{res}}(s)$ to denote the sub-optimality bounds in (28) when using the action set $A(s)$ from (30) and (31), respectively.

We compare the sample-path bound in (29) with the infinite-horizon sup-norm and weighted-norm bounds of [14]. Because the distance between the finite- and discounted infinite-horizon objectives is bounded by $\gamma^T \|c\|_\infty / (1 - \gamma)$, the value functions at $t = 1$ agree to numerical precision (at $T = 100$ and $\gamma = 0.75$, $\gamma^T < 10^{-12}$). For both models, we compute V_t^* , $\hat{V}_t^*$, π_t^* , $\hat{\pi}_t^*$ via backward induction, and $V_1^{\hat{\pi}^*}$ via policy evaluation. All three bounds are plotted in Fig. 1a (full state space) and Fig. 1b (zoomed into the region $\{-10, \dots, 10\}$). The shaded region between $V_1^{\hat{\pi}^*}(s)$ and the tightest lower bound indicates where $V_1^*(s)$ must lie.

The sample-path bound using $2\alpha_1^{\text{res}}(s)$ is substantially tighter than both existing bounds in the operating region. At $s = 3$, the sample-path bound gives $2\alpha_1^{\text{res}}(3) = 42.31$ while the weighted-norm bound gives 62.25; at $s = 10$, the values are 33.78 versus 83.46. Both are well below the sup-norm bound of 891.20.

Fig. 2 compares $2\alpha_1^{\max}(s)$ with $2\alpha_1^{\text{res}}(s)$. As can be seen from the figure, bounds are very close, confirming that the restricted action set captures most of the tightening without requiring computation of the true optimal policy.

¹Ordering above the maximum demand is never optimal, as it guarantees positive inventory without reducing shortage risk

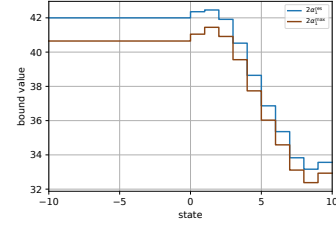


Fig. 2: Comparison of $2\alpha_1^{\max}(s)$ and $2\alpha_1^{\text{res}}(s)$ in the region $s \in \{-10, \dots, 10\}$.

C. Observations over a packet drop network

Consider the POMDP model of Section II where the controller receives observations over a packet drop network. Specifically, the observation Y_t is given as

$$Y_t = \begin{cases} \mathfrak{E} & \text{if packet drop occurs} \\ S_t & \text{otherwise} \end{cases}$$

where $\mathfrak{E}$ denotes packet loss. We assume that at each time t , the observation packet is dropped with probability p independent of the past history of the network or the POMDP. We further assume that at each time t , $c_t(h_t, a_t) = \int k_t(s_t, a_t) P(ds_t | h_t)$ where $0 \leq k_t(s_t, a_t) \leq k_{\max}$.

We consider the following AIS generator for the POMDP (see Definition 3): $\mathcal{Z} = \mathcal{S}$ and

•

$$\sigma_t(y_{1:t}, a_{1:t-1}) = \begin{cases} y_t & \text{if } y_t \neq \mathfrak{E} \\ s_{\text{nom}} & \text{otherwise} \end{cases}$$

where $s_{\text{nom}} \in \mathcal{S}$ is nominal state (e.g. 0).

• $\hat{c}_t(s, a) = k_t(s, a)$.

• $\hat{P}_t = P_{S,t}$ where $P_{S,t}$ is the state transition kernel of $\mathcal{P}$.

With the POMDP model and the AIS generator described above, it can be verified that Assumptions 2 and 4 (with Total variation as the IPM) are satisfied with the following $\varepsilon_t(h_t, a)$ and $\delta_t(h_t, a)$:

$$\begin{aligned} \varepsilon_t(h_t, a) &= \varepsilon_t(y_{1:t}, a_{1:t-1}, a) = k_{\max} \mathbb{1}_{\{y_t = \mathfrak{E}\}}, \\ \delta_t(h_t, a) &= \delta_t(y_{1:t}, a_{1:t-1}, a) = 2 \mathbb{1}_{\{y_t = \mathfrak{E}\}}. \end{aligned} \quad (32)$$

In order to identify the α_t, β_t functions of Theorem 2, we first define the following sequence of real numbers:

$$\xi_{T-1} = p k_{\max}, \quad (33)$$

and for $t < T - 1$,

$$\xi_t = \xi_{t+1} + p(k_{\max} + \text{span}(\hat{V}_{t+1}^*)). \quad (34)$$

We can now use Theorem 2 to obtain the following sub-optimality bound (proof omitted due to space limit).

Corollary 5 For all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, define $\alpha_t(h_t)$ and $\beta_t(h_t, a_t)$ as follows:

$$\beta_T(y_{1:T}, a_{1:T}) = \alpha_T(y_{1:T}, a_{1:T-1}) = k_{\max} \mathbb{1}_{\{y_T = \mathfrak{E}\}}, \quad (35)$$

$$\begin{aligned} \beta_{T-1}(y_{1:T-1}, a_{1:T-1}) &= \alpha_{T-1}(y_{1:T-1}, a_{1:T-2}) \\ &= \xi_{T-1} + (k_{\max} + \text{span}(\hat{V}_T^*)) \mathbb{1}_{\{y_{T-1} = \mathfrak{E}\}} \end{aligned} \quad (36)$$

and for $t < T - 1$:

$$\begin{aligned}\beta_t(y_{1:t}, a_{1:t}) &= \alpha_t(y_{1:t}, a_{1:t-1}) \\ &= \xi_t + (k_{\max} + \text{span}(\hat{V}_{t+1}^*)) \mathbb{1}_{\{y_t = \mathfrak{E}\}}.\end{aligned}\quad (37)$$

Then, $V_t^\pi(h_t) - V_t^*(h_t) \leq 2\alpha_t(h_t)$.

Remark 4 Using the result of Corollary 5 at $t = 1$ and averaging over the history $H_1 = Y_1$ at time $t = 1$, we get the following performance bound between the total expected cost of policy π and the optimal total expected cost:

$$E[V_1^\pi(H_1)] - E[V_1^*(H_1)] \leq 2\xi_1 + 2p(k_{\max} + \text{span}(\hat{V}_2^*)). \quad (38)$$

If we use Theorem 1 for the example described above, the value function sub-optimality gap evaluates to $V_t^\pi(h_t) - V_t^*(h_t) \leq 2\bar{\alpha}_t$ where $\bar{\alpha}_T = k_{\max}$ and for $t < T$,

$$\bar{\alpha}_t = (k_{\max} + \text{span}(\hat{V}_{t+1}^*)) + \bar{\alpha}_{t+1},$$

which implies that

$$E[V_1^\pi(H_1)] - E[V_1^*(H_1)] \leq 2\bar{\alpha}_1. \quad (39)$$

Note that the performance bound in (38) approaches 0 as $p \rightarrow 0$ whereas the performance bound in (39) does not.

V. CONCLUSION

We have developed sample path dependent sub-optimality bounds for approximate information state-based policies. By allowing the approximation errors to depend on the realized history and action, our bounds replace worst-case constants with pointwise error terms and their expectations under the trajectory distribution, yielding tighter performance guarantees. The error is characterized through an α - β recursion where $\beta_t(h_t, a_t)$ captures action-specific Q-function approximation errors and $\alpha_t(h_t)$ captures value function approximation errors. We established these bounds for both centralized and decentralized (restricted policy class) settings.

Several directions for future work are worth pursuing. First, developing practical methods to estimate or bound $\alpha_t(h_t)$ for specific problem classes would make the bounds actionable. The fictitious control problem interpretation suggests a hierarchical approach where the original AIS framework is applied at a meta-level to bound the sample path dependent errors. Second, extending these results to the infinite-horizon discounted setting would broaden their applicability to reinforcement learning problems.

APPENDIX

We will prove the theorem by proving the following lemma by backward induction.

Lemma 2 Suppose that for each $t \in \{1, \dots, T\}$, there exist measurable functions $\alpha_t: \mathcal{H}_t \rightarrow \mathbb{R}$ and $\beta_t: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ that satisfy the conditions of Theorem 2. Then, under Assumptions 2 and 3, for each $t \in \{1, \dots, T\}$ and each $h_t \in \mathcal{H}_t$,

A. Q-function bound

$$\begin{aligned}|Q_t^*(h_t, a) - \hat{Q}_t^*(\sigma_t(h_t), a)| &\leq \beta_t(h_t, a), \\ |Q_t^\pi(h_t, a) - \hat{Q}_t^\pi(\sigma_t(h_t), a)| &\leq \beta_t(h_t, a).\end{aligned}$$

B. Value function approximation bounds

$$\begin{aligned}|V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t))| &\leq \alpha_t(h_t), \\ |V_t^\pi(h_t) - \hat{V}_t^\pi(\sigma_t(h_t))| &\leq \alpha_t(h_t).\end{aligned}$$

C. Sub-optimality bound

$$|V_t^\pi(h_t) - V_t^*(h_t)| \leq 2\alpha_t(h_t).$$

We now prove the lemma starting backward at $t = T$.

Base case ($t = T$). At the terminal time, $Q_T^*(h_T, a) = c_T(h_T, a)$ and $\hat{Q}_T^*(z_T, a) = \hat{Q}_T^\pi(z_T, a) = \hat{c}_T(z_T, a)$.

Step A: Q-function bound. By Assumption 2:

$$\begin{aligned}|Q_T^*(h_T, a_T) - \hat{Q}_T^*(\sigma_T(h_T), a_T)| \\ = |Q_T^\pi(h_T, a_T) - \hat{Q}_T^\pi(\sigma_T(h_T), a_T)| \\ \leq \varepsilon_T(h_T, a_T) = \beta_T(h_T, a_T).\end{aligned}$$

Step B: Value function approximation bounds. We establish one-sided bounds on $V_T^* - \hat{V}_T^*$ and $\hat{V}_T^* - V_T^*$ separately.

For the first direction:

$$\begin{aligned}V_T^*(h_T) - \hat{V}_T^*(\sigma_T(h_T)) \\ = Q_T^*(h_T, \pi_T^*(h_T)) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) \\ \leq Q_T(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) \\ \leq \beta_T(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \leq \alpha_T(h_T)\end{aligned}\quad (40)$$

where the first inequality uses $Q_T^*(h_T, \pi_T^*(h_T)) \leq Q_T^*(h_T, a)$ for any a (optimality of π_T^*).

For the reverse direction:

$$\begin{aligned}\hat{V}_T^*(\sigma_T(h_T)) - V_T^*(h_T) \\ = \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) - Q_T^*(h_T, \pi_T^*(h_T)) \\ \leq \hat{Q}_T^*(\sigma_T(h_T), \pi_T^*(h_T)) - Q_T^*(h_T, \pi_T^*(h_T)) \\ \leq \beta_T(h_T, \pi_T^*(h_T)) \leq \alpha_T(h_T).\end{aligned}\quad (41)$$

Combining (40) and (41) gives

$$|V_T^*(h_T) - \hat{V}_T^*(\sigma_T(h_T))| \leq \alpha_T(h_T).$$

For the policy value functions, since $\pi = \hat{\pi}^* \circ \sigma$:

$$\begin{aligned}|V_T^\pi(h_T) - \hat{V}_T^\pi(\sigma_T(h_T))| \\ = |Q_T^\pi(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T)))| \\ \leq \beta_T(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \leq \alpha_T(h_T).\end{aligned}\quad (42)$$

Step C: Sub-optimality bound. By triangle inequality:

$$\begin{aligned}|V_T^\pi(h_T) - V_T^*(h_T)| \\ \leq |V_T^\pi(h_T) - \hat{V}_T^\pi(\sigma_T(h_T))| + |\hat{V}_T^\pi(\sigma_T(h_T)) - V_T^*(h_T)| \\ \leq \alpha_T(h_T) + \alpha_T(h_T) = 2\alpha_T(h_T).\end{aligned}$$

Induction step. Assuming the bounds of Lemma 2 holds at time $t + 1$, then we proceed to prove them at time t .

Step A: Q-function bound. We decompose the Q-function approximation error into reward and future-value terms:

$$\begin{aligned} & |Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t)| \\ & \leq \underbrace{\left| \int c_t(s_t, a_t) P(ds_t|h_t) - \hat{c}_t(\sigma_t(h_t), a_t) \right|}_{\leq \varepsilon_t(h_t, a_t) \text{ by Assumption 2}} \\ & \quad + \left| \int P(dy_{t+1}|h_t, a_t) V_{t+1}^*(h_t, a_t, y_{t+1}) \right. \\ & \quad \left. - \int \hat{P}(dz_{t+1}|z_t, a_t) \hat{V}_{t+1}^*(z_{t+1}) \right|. \end{aligned} \quad (43)$$

For the future-value term, we add and subtract $\int P(dy_{t+1}|h_t, a_t) \hat{V}_{t+1}^*(\sigma_{t+1}(h_t, a_t, y_{t+1}))$ to obtain:

$$\begin{aligned} \text{2nd term} & \leq \left| V_{t+1}^*(h_t, a_t, y_{t+1}) \right. \\ & \quad \left. - \hat{V}_{t+1}^*(\sigma_{t+1}(h_t, a_t, y_{t+1})) \right| P(dy_{t+1}|h_t, a_t) \\ & \quad + \left| \int \hat{V}_{t+1}^*(z_{t+1}) \nu_t(dz_{t+1}|h_t, a_t) \right. \\ & \quad \left. - \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}(dz_{t+1}|z_t, a_t) \right|, \end{aligned} \quad (44)$$

The first term in (44) is bounded, using the induction hypothesis, by $\int \alpha_{t+1}(h_t, a_t, y_{t+1}) P(dy_{t+1}|h_t, a_t)$. The second term is bounded by $\Delta_t(h_t, a_t)$ by Assumption 3. Combining the above we get the Q-function bound:

$$\begin{aligned} & |Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t)| \leq \varepsilon_t(h_t, a_t) \\ & \quad + \int \alpha_{t+1}(h_t, a_t, y_{t+1}) P(dy_{t+1}|h_t, a_t) + \Delta_t(h_t, a_t) \\ & \leq \beta_t(h_t, a_t), \end{aligned} \quad (45)$$

where the last inequality is the property of $\beta(h_t, a_t)$ from (8).

A similar argument as above by replacing Q^* and V^* with Q^π and V^π gives us $|Q_t^\pi(h_t, a) - \hat{Q}_t^*(\sigma_t(h_t), a)| \leq \beta_t(h_t, a)$.

Step B: Value function approximation bounds. Similar to the base case, we consider the two one-sided bounds separately. For the first direction, from the above established Q-function bound we have the following:

$$\begin{aligned} & V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ & = Q_t^*(h_t, \pi_t^*(h_t)) - \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) \\ & \leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) \\ & \leq \beta_t(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \leq \alpha_t(h_t). \end{aligned}$$

For the reverse direction, applying the Q-function bound again we obtain:

$$\begin{aligned} & \hat{V}_t^*(\sigma_t(h_t)) - V_t^*(h_t) \\ & = \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) - Q_t^*(h_t, \pi_t^*(h_t)) \\ & \leq \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t)) - Q_t^*(h_t, \pi_t^*(h_t)) \\ & \leq \beta_t(h_t, \pi_t^*(h_t)) \leq \alpha_t(h_t). \end{aligned}$$

Hence $|V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t))| \leq \alpha_t(h_t)$.

For the policy value functions, the induction step gives

$$\begin{aligned} & |V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t))| \\ & = |Q_t^\pi(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t)))| \\ & \leq \beta_t(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \leq \alpha_t(h_t). \end{aligned}$$

Note that the two one-sided bounds on $V_t^* - \hat{V}_t^*$ show that $\alpha_t(h_t)$ needs only to dominate β_t at the two actions $\pi_t^*(h_t)$ and $\hat{\pi}_t^*(\sigma_t(h_t))$, which justifies the characterization (9).

Step C: Sub-optimality bound. By triangle inequality:

$$\begin{aligned} & |V_t^\pi(h_t) - V_t^*(h_t)| \\ & \leq |V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t))| + |\hat{V}_t^*(\sigma_t(h_t)) - V_t^*(h_t)| \\ & \leq \alpha_t(h_t) + \alpha_t(h_t) = 2\alpha_t(h_t). \end{aligned}$$

This completes the induction step.

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